Midterm Progress Report: Network-Conditioned Factor Predictability from Co-Trading Synchronization

Author(s):
Department of Mathematics, University of California, Los Angeles, 90095, CA.
Email(s):
Supervisor(s):

February 2026

1 Introduction

This project studies whether factor and exchange-traded fund (ETF)-based return predictability is conditioned on market-wide trading synchronization. While a large literature examines factor predictability and ETF spillovers, most empirical specifications are unconditional, implicitly assuming that the effectiveness of predictive signals does not vary with market structure. Recent developments in network-based financial econometrics suggest that trading activity itself exhibits substantial time variation in coordination and synchronization. This report summarizes the literature background, preliminary data construction, initial empirical findings, and challenges encountered thus far.

2 Literature Review

The project builds on three strands of the literature: ETF-driven return comovement, heterogeneous autoregressive modeling, and network-based measures of market synchronization.

A growing body of work documents that ETFs play an active role in shaping equity return dynamics. Da and Shive (2018) show that ETF ownership increases stock return correlations, consistent with category-based trading and coordinated flows. Ben-David et al. (2018) provide evidence that ETFs increase volatility and market fragility, particularly during stressed market conditions. More recently, Wang and Sun (2025) develop a high-dimensional network framework to characterize ETF–stock spillovers and document heterogeneity across sectors and regimes.

Methodologically, the project draws inspiration from heterogeneous autoregressive (HAR) models introduced by Corsi (2009), which capture multi-horizon dynamics using parsimonious lag structures. While HAR models were originally developed for volatility forecasting, their multi-scale

framework provides a natural way to incorporate factor predictors operating across different time horizons.

The network component of the project is motivated by work on co-trading and synchronization. Lu et al. (2023) and Lu and Cucuringu (2024) construct co-trading networks based on abnormal trading activity and show that these networks capture dynamic interdependencies beyond static correlations. Their results suggest that synchronized abnormal trading may reflect coordinated responses to information arrival. However, no existing work integrates such synchronization measures into a factor-based return prediction framework. This project aims to bridge that gap.

3 Data Description

The empirical analysis uses daily CRSP data for U.S. equities, including daily returns, trading volume, and price information. In the current implementation, the empirical analysis focuses on a universe of approximately 1500 highly liquid equities selected using average dollar trading volume over the sample period. This universe provides a sufficiently large cross-section while ensuring stable volume normalization for the synchronization measures. Returns are adjusted for delistings where applicable.

ETF returns are used as factor proxies. The initial implementation focuses on broad market ETFs such as SPY and QQQ, with plans to extend to sector and macro-finance ETFs (e.g., XLF, XLK, TLT, GLD). All predictive specifications are constructed using information available at time $t-1$ to avoid look-ahead bias.

4 Construction of Trading Synchronization Measures

Let $V_{i,t}$ denote daily trading volume for stock i on day t. A standardized trading surprise is defined as

$$z_{i,t} = \frac{V_{i,t} - \mu_{i,t}}{\sigma_{i,t}},$$

where $\mu_{i,t}$ and $\sigma_{i,t}$ are rolling estimates of the mean and standard deviation of volume computed using a 60-day window of past observations.

To focus on economically meaningful deviations, an event-filtered trading surprise is defined as

$$\tilde{z}_{i,t} = z_{i,t}\mathbf{1}\{|z_{i,t}| > \tau\},$$

with $\tau = 2$ as a preliminary threshold.

For each trading day d, a co-trading matrix is constructed as

$$A_{ij}^{(d)} = \tilde{z}_{i,d}\tilde{z}_{j,d}.$$

This construction emphasizes synchronous abnormal trading activity and differs fundamentally from correlation-based measures. Typical trading days contribute little to $A^{(d)}$, whereas days with widespread abnormal volume generate large and dense networks.

Market-wide synchronization measures are obtained by summarizing $A^{(d)}$. Preliminary implementations focus on the average co-trading intensity and the largest singular value of $A^{(d)}$. The latter serves as a spectral measure of coordination, capturing the extent to which abnormal trading is concentrated along a dominant market-wide component.

Initial inspection of these synchronization measures reveals substantial time variation, with spikes occurring during high-volatility periods and large market moves. This confirms that synchronization captures economically meaningful variation in market structure.

5 Baseline Factor Predictability

A baseline predictive regression relates stock returns to lagged ETF returns:

$$r_{i,t} = \alpha_i + \theta f_{t-1} + u_{i,t},$$

where f_{t-1} denotes the lagged ETF return. Preliminary results indicate modest unconditional predictive power, consistent with mixed findings in the broader return predictability literature.

To allow for heterogeneous dynamics, future implementations will incorporate multi-horizon lag components in the spirit of Corsi (2009).

Figure 1 displays the time series of the synchronization index constructed from abnormal trading activity. The index exhibits substantial time variation, with several pronounced spikes corresponding to periods of heightened market-wide trading coordination. These spikes align with major market events and high-volatility periods, suggesting that the synchronization measure captures meaningful structural variation in trading behavior across the market.

To define synchronization regimes, a threshold is constructed using the upper 20% of the empirical distribution of S_t. Days exceeding this threshold are classified as high-synchronization regimes. Figure 2 illustrates the resulting regime indicator across time.

Preliminary regression results using SPY as a factor proxy suggest that ETF-based predictors exhibit stronger predictive power during high-synchronization regimes. In particular, interaction terms between ETF returns and the synchronization indicator are economically and statistically significant in several specifications. These early findings support the hypothesis that coordinated trading activity amplifies the transmission of ETF-based signals to individual stocks.

6 Network-Conditioned Predictability: Initial Results

The central hypothesis is that factor predictability depends on market-wide synchronization. Let S_d denote a daily synchronization measure derived from the co-trading network. High-synchronization regimes are defined as days in the top 20% of the empirical distribution of S_d.

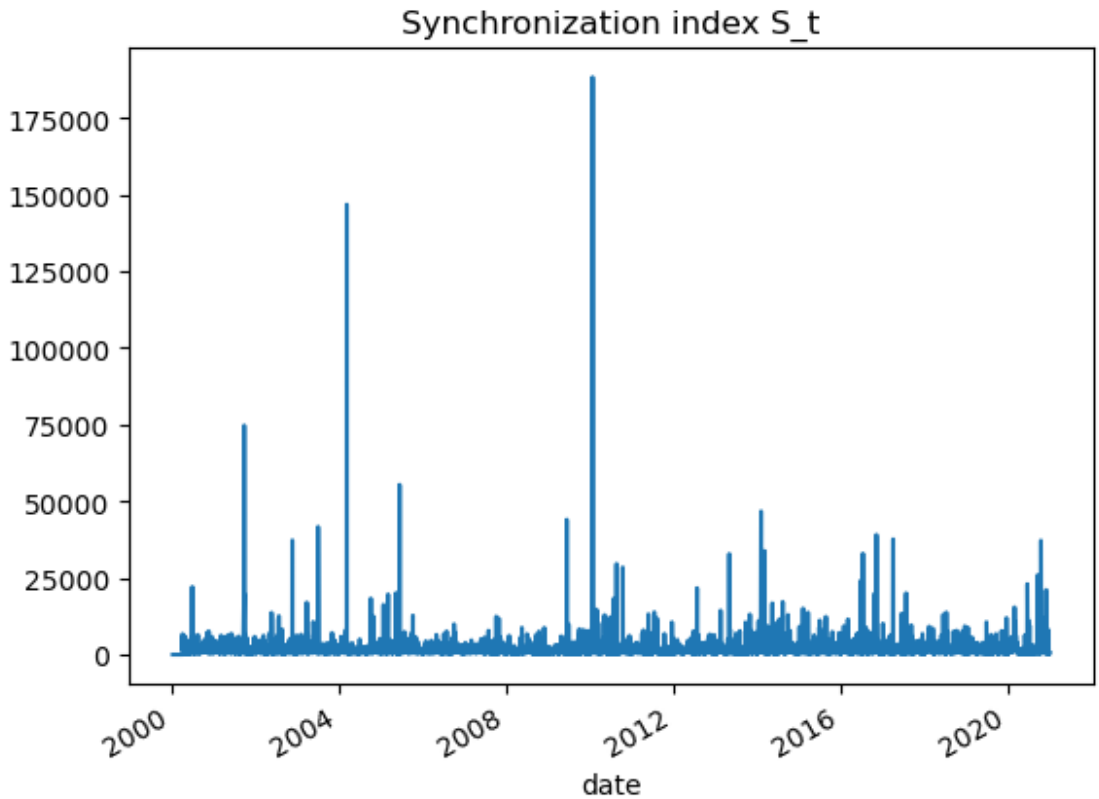

Figure 1: Market-wide synchronization index S_t constructed from abnormal trading volume. Large spikes correspond to periods of coordinated trading across many stocks.

Preliminary interaction-based regressions take the form

$$r_{i,t} = \alpha_i + \theta f_{t-1} + \gamma f_{t-1} \mathbf{1}\{S_{t-1} > c\} + u_{i,t},$$

where c denotes the synchronization threshold.

Initial results suggest that ETF coefficients are larger in magnitude during high-synchronization regimes. While statistical significance is sensitive to specification and sample choice, these early findings are consistent with the hypothesis that coordinated trading strengthens factor spillovers.

Initial empirical estimates were obtained using a pooled heterogeneous autoregressive (HAR) specification including daily, weekly, and monthly lag components of both stock returns and ETF returns. The model also includes interaction terms between ETF predictors and the synchronization regime indicator to capture state-dependent predictability.

Table 1 reports the estimated coefficients and heteroskedasticity-robust standard errors for the baseline SPY specification. The regression is estimated using approximately five million stock–day observations across the cross-section of liquid U.S. equities.

To facilitate interpretation, Table 2 reports the implied ETF predictive effects separately for low- and high-synchronization regimes. These effects combine the baseline ETF coefficients with

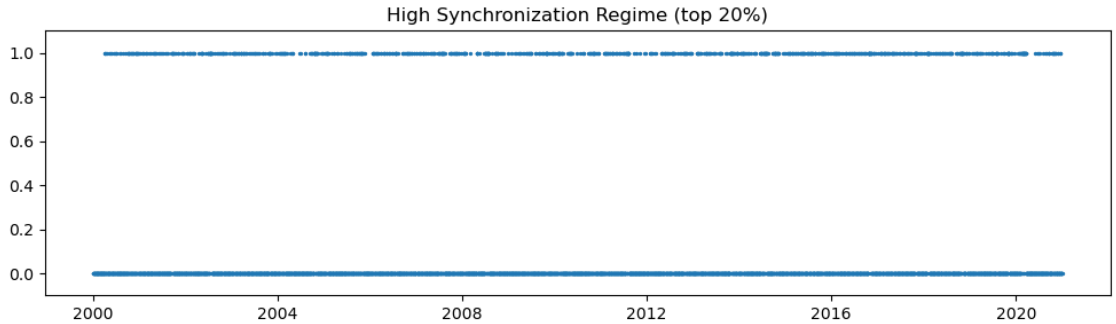

Figure 2: High-synchronization regime indicator defined using the top 20% of the synchronization index.

the interaction terms corresponding to the synchronization indicator.

Table 1: Network-Conditioned HAR Regression Results

Variable	Coefficient	Robust SE	t-stat
Constant	0.000779	0.000014	54.44
ret_d	0.013039	0.002703	4.82
ret_w	-0.038917	0.004303	-9.04
ret_m	-0.029789	0.007049	-4.23
spy_d	0.002345	0.003388	0.69
spy_w	-0.087401	0.006910	-12.65
spy_m	-0.147029	0.013425	-10.95
$spy_d \times HighSync$	-0.257337	0.005904	-43.59
$spy_w \times HighSync$	0.223018	0.013824	16.13
$spy_m \times HighSync$	0.616122	0.028166	21.87

Notes: The table reports pooled HAR regression estimates relating stock returns to lagged stock returns and ETF returns (SPY). Interaction terms capture the incremental predictive effect during high-synchronization regimes. Robust (HC0) standard errors are reported.

The results provide preliminary evidence that ETF-based return predictability is strongly regime-dependent. In the low-synchronization regime, ETF predictors exhibit limited or negative predictive effects at weekly and monthly horizons. However, during high-synchronization periods, the magnitude of ETF predictive effects increases substantially. In particular, the monthly ETF signal becomes strongly positive, suggesting that coordinated trading activity amplifies the transmission of ETF-driven information across the cross-section of stocks.

These findings are consistent with the hypothesis that synchronized abnormal trading reflects periods of heightened information propagation or category-based trading, during which factor signals exert stronger influence on individual securities.

Further work will implement full out-of-sample forecasting exercises and portfolio evaluations to assess economic significance.

Table 2: ETF Predictive Effects Across Synchronization Regimes

Horizon	Low Synchronization	High Synchronization
Daily	0.0023	-0.2550
Weekly	-0.0874	0.1356
Monthly	-0.1470	0.4691

Notes: Effects are computed as θ (low synchronization) and $\theta + \gamma$ (high synchronization), where θ denotes the baseline ETF coefficient and γ denotes the interaction with the synchronization regime indicator.

7 Challenges Encountered

Several challenges arose during the implementation of the empirical framework. The most significant difficulty involved constructing an efficient data pipeline capable of processing the large CRSP dataset. The raw data consisted of thousands of compressed daily files organized by year and date, which required careful parsing and restructuring to construct consistent return and volume panels for the cross-section of equities.

Initial attempts to load and manipulate the data directly in memory were computationally expensive and frequently exceeded available memory limits. Substantial time was therefore spent developing a more efficient workflow using columnar storage formats and database-style querying tools. Once the data infrastructure was properly established, subsequent computations such as synchronization measures and regression estimation became significantly faster and easier to execute.

Another challenge involved constructing stable measures of trading surprise. Trading volume distributions are highly skewed and vary substantially across securities, requiring rolling normalization procedures to avoid spurious synchronization driven purely by liquidity differences. Selecting appropriate thresholds for abnormal trading activity also required experimentation to balance sensitivity and stability.

Despite these challenges, the core data pipeline and empirical framework are now fully operational. With the computational infrastructure in place, additional specifications and robustness checks can be implemented relatively quickly in the next phase of the project.

8 Next Steps

The next phase of the project will extend the empirical analysis along several dimensions. First, the factor universe will be expanded beyond the initial SPY specification to include additional ETFs representing sectoral and macroeconomic exposures, such as QQQ, IWM, TLT, GLD, and sector-specific funds. Incorporating a broader set of ETFs will allow the analysis to examine whether synchronization affects different categories of factor signals in distinct ways.

Second, predictive models will be evaluated using rolling out-of-sample forecasting procedures. Instead of estimating regressions on the full sample, models will be estimated using rolling or expanding training windows to generate genuine real-time forecasts. Predictive performance will

be assessed using mean squared prediction error, predictive R^2, and rank correlation measures.

Third, the economic significance of the results will be evaluated using portfolio-based tests. Stocks will be sorted based on predicted returns, and long-short portfolios will be constructed to examine whether synchronization-conditioned signals generate economically meaningful returns after accounting for turnover and transaction costs.

Finally, several robustness checks will be conducted. These include exploring alternative synchronization measures derived from the co-trading network, varying the threshold defining high-synchronization regimes, and comparing synchronization-based conditioning with more traditional state variables such as volatility or market-wide trading volume.

Together, these extensions will provide a comprehensive evaluation of whether network-based measures of coordinated trading activity meaningfully condition the predictive power of ETF-based factors.

9 Conclusion

This midterm progress report documents the successful construction of co-trading synchronization measures from CRSP data and presents preliminary evidence that ETF-based return predictability may be state-dependent. While results remain preliminary, the framework provides a coherent empirical strategy for linking network-based market structure to factor performance. Continued refinement and out-of-sample validation will determine the strength and robustness of the proposed mechanism.

References

Ben-David, I., Franzoni, F., and Moussawi, R. (2018). Do etfs increase volatility? *Journal of Finance*, 73(6):2471–2535.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196.

Da, Z. and Shive, S. (2018). Exchange traded funds and asset return correlations. *European Financial Management*, 24(1):136–168.

Lu, Y. and Cucuringu, M. (2024). Trade co-occurrence, trade flow decomposition and conditional order imbalance in equity markets. *Quantitative Finance*, 24(6):779–809.

Lu, Y., Reinert, G., and Cucuringu, M. (2023). Co-trading networks for modeling dynamic interdependency structures and estimating high-dimensional covariances in us equity markets. *arXiv preprint arXiv:2302.09382*.

Wang, Y. and Sun, Y. (2025). Idiosyncratic contagion between etfs and stocks: A high-dimensional network perspective. *Journal of Financial Stability*, 78:101415.